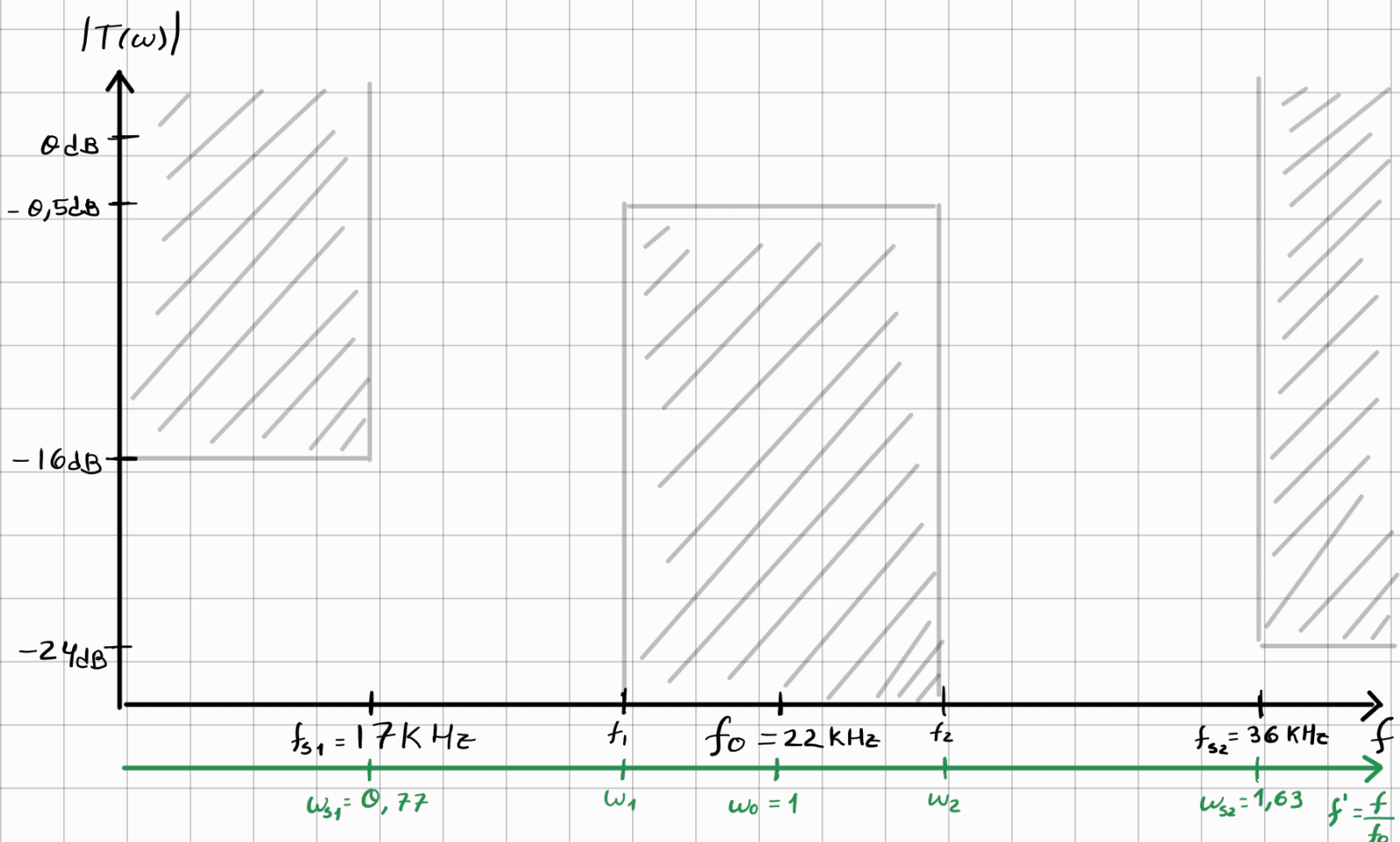


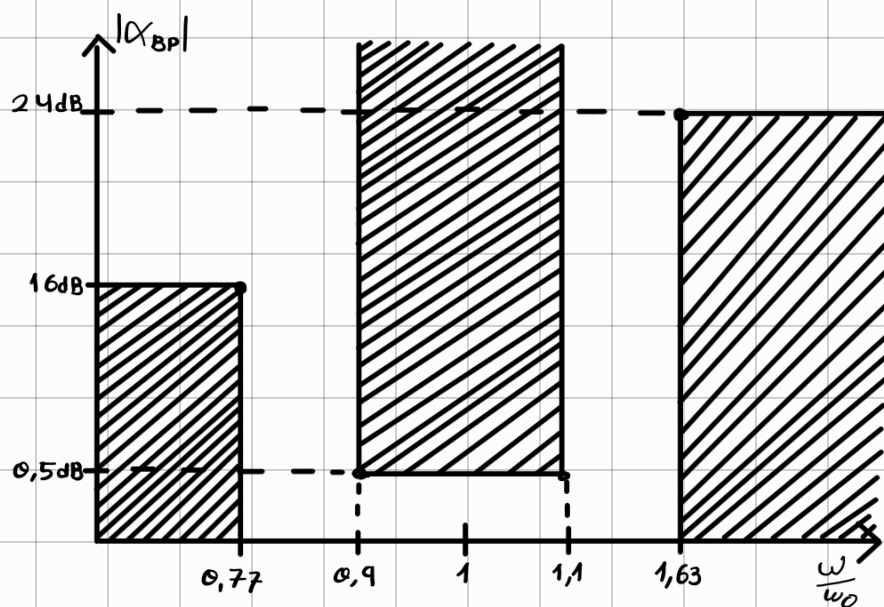
1)



- $B = \frac{\omega_0}{Q} = 0,2$

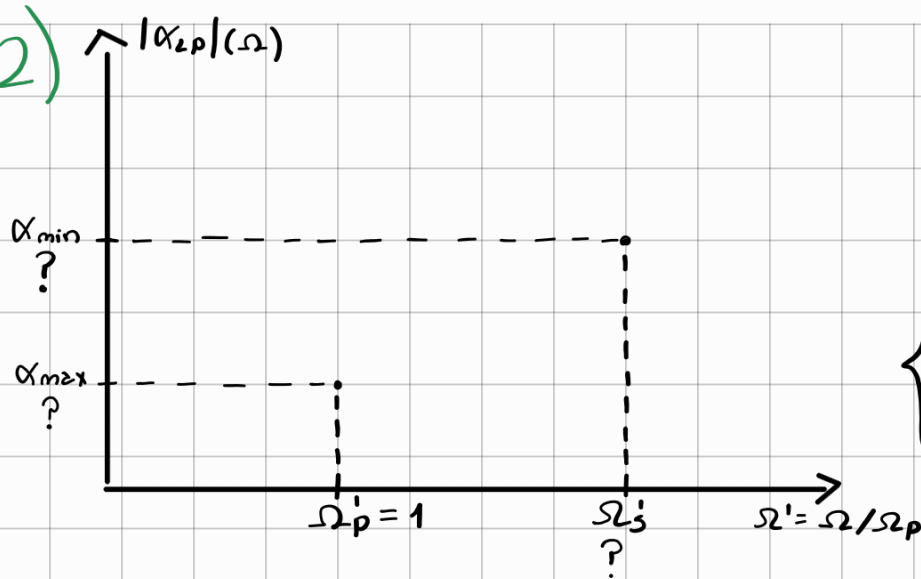
- $\begin{cases} \omega_1 \cdot \omega_2 = \omega_0^2 = 1 \\ \omega_2 - \omega_1 = B = 0,2 \end{cases}$

$$\Rightarrow \begin{cases} \omega_1 = 0,9 \\ \omega_2 = 1,1 \end{cases}$$



$$\begin{cases} X_{\max} = 0,5 \text{ dB} \\ X_{\min_1} = 16 \text{ dB} \\ X_{\min_2} = 24 \text{ dB} \end{cases}$$

2)



$$\Omega(\omega) = \frac{1}{B_{BP}} \cdot \frac{\omega^2 - \omega_{0BP}^2}{\omega}$$

$$\begin{cases} \Omega_{s1} = \Omega(\omega_{s1}) = -2,64 / \alpha_{min_1} = 16 \text{ dB} \\ \Omega_{s2} = \Omega(\omega_{s2}) = 5,08 / \alpha_{min_2} = 24 \text{ dB} \end{cases}$$

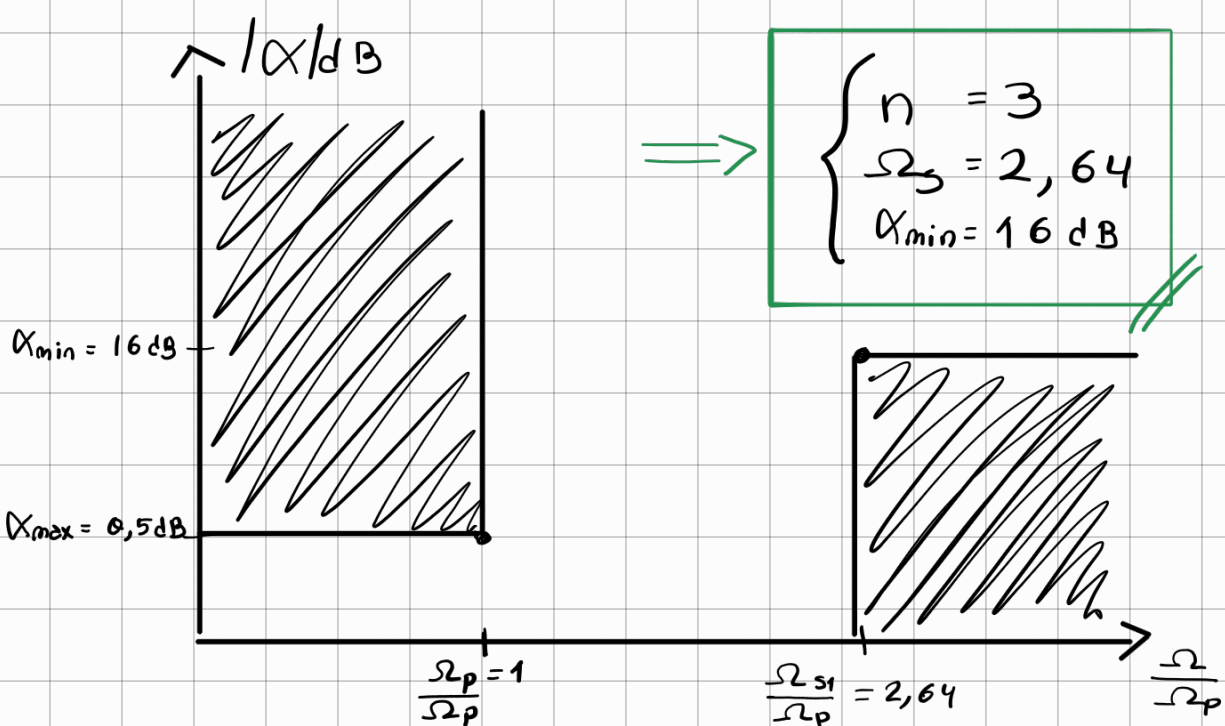
$$\epsilon^2 = 10^{\left(\frac{\alpha_{max}}{10}\right)} - 1 \longrightarrow \epsilon^2 = 0,122 //$$

$$\alpha_{min} = 10 \log \left[1 + \epsilon^2 \left(\cosh^2(n \cdot \text{arccosh}(\Omega_s)) \right) \right]$$

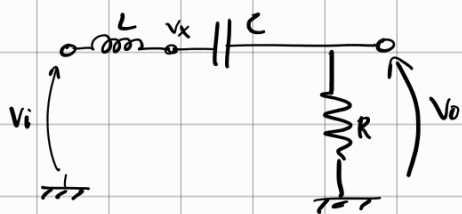
n_1	$\alpha_{min_1} = 16 \text{ dB}$
1	2,67 dB
2	13,31 dB
3	27,22 dB

n_2	$\alpha_{min_2} = 24 \text{ dB}$
1	6,18 dB
2	24,96 dB
3	45 dB

Nos quedamos con el mayor orden



Bonus: Pass Banda



Filtro Pass Banda con
una R_G de atenuación
(pasivo)

- $(V_x - V_i) \cdot \frac{1}{sL} + (V_x - V_o) sC = 0$

- $(V_o - V_x) sC + V_o \frac{1}{R} = 0$

$\rightarrow V_x = V_o \frac{s + \frac{1}{CR}}{s}$

$\rightarrow V_x \left(\frac{1}{sL} + sC \right) - V_o sC = \frac{V_i}{sL}$
 $V_o \left[\frac{s + \frac{1}{CR}}{s} \left(\frac{1}{sL} + sC \right) - sC \right] = \frac{V_i}{sL}$
 $\frac{V_o}{V_i} \left[\frac{s^2 \frac{L}{R} + s + \frac{1}{CR}}{s^2 L} \right] = \frac{1}{sL} \quad \frac{\frac{R}{L} s}{s^2 + s \frac{R}{L} + \frac{1}{LC}}$

$$\frac{V_o}{V_i} = \frac{s \frac{R}{L}}{s^2 + s \frac{R}{L} + \frac{1}{LC}}$$

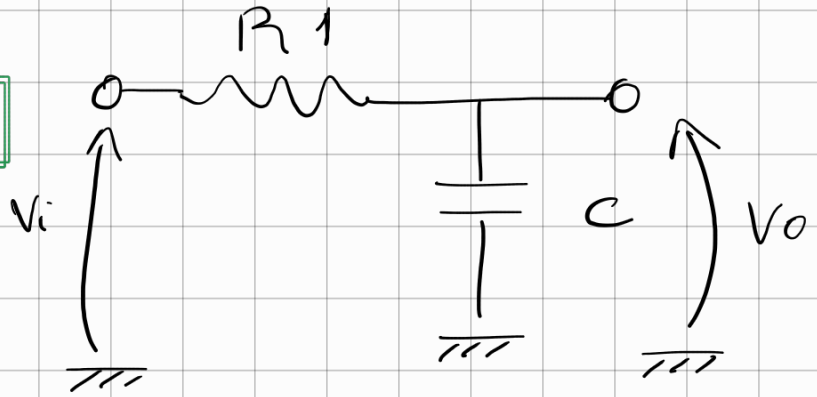
$$\left\{ \begin{array}{l} \omega_0 = \frac{1}{\sqrt{LC}} \longrightarrow \boxed{L = \frac{1}{\omega_0^2 \cdot C}} \\ Q = \frac{1}{R} \cdot \sqrt{\frac{L}{C}} \longrightarrow \boxed{R = \frac{1}{Q \omega_0 C}} \end{array} \right.$$

Bonus: Pass Bajos

$$\frac{V_o}{V_i} = \frac{\frac{1}{sC}}{R1 + \frac{1}{sC}} = \frac{1}{sCR1 + 1} = \boxed{\frac{\frac{1}{CR1}}{s + \frac{1}{CR1}}}$$

$$\hookrightarrow \sigma = \frac{1}{C \cdot R1} = 0,6265$$

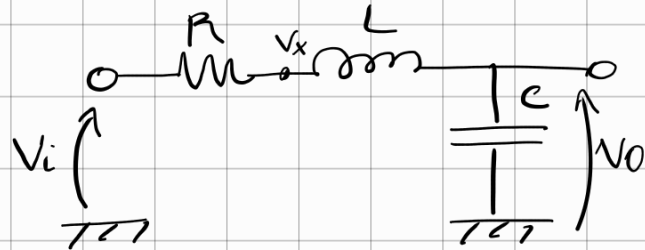
$$\Rightarrow R1 = \frac{1}{\sigma \cdot C}$$



$$\bullet (V_x - V_i) \frac{1}{R} + (V_x - V_o) \frac{1}{sL} = 0$$

$$\bullet (V_o - V_x) \frac{1}{sL} + V_o sC = 0$$

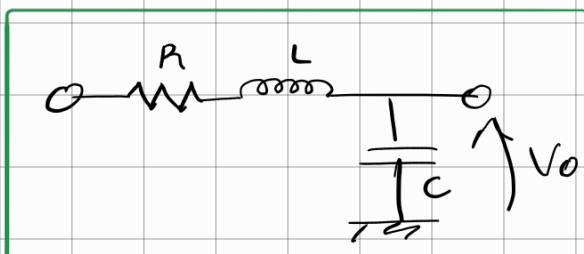
$$\hookrightarrow (1 + s^2 LC) V_o = V_x$$



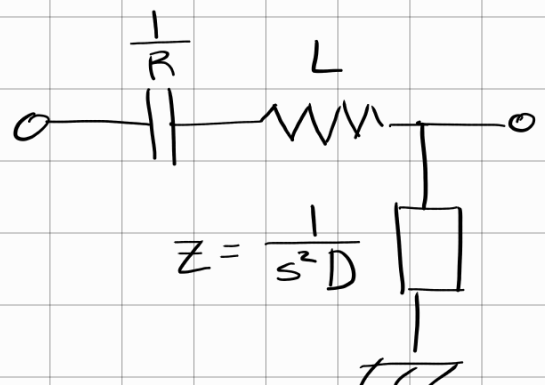
$$\hookrightarrow \frac{V_o}{V_i} \left[(s^2 LC + 1) \left(\frac{1}{sL} + \frac{1}{R} \right) - \frac{1}{sL} \right] = \frac{1}{R} \Rightarrow \boxed{\frac{V_o}{V_i} = \frac{\frac{1}{LC}}{s^2 + s\frac{R}{L} + \frac{1}{LC}}}$$

$$\left\{ \begin{array}{l} \omega_0 = \frac{1}{LC} = 1,069 \Rightarrow L = \frac{1}{\omega_0 C} \end{array} \right.$$

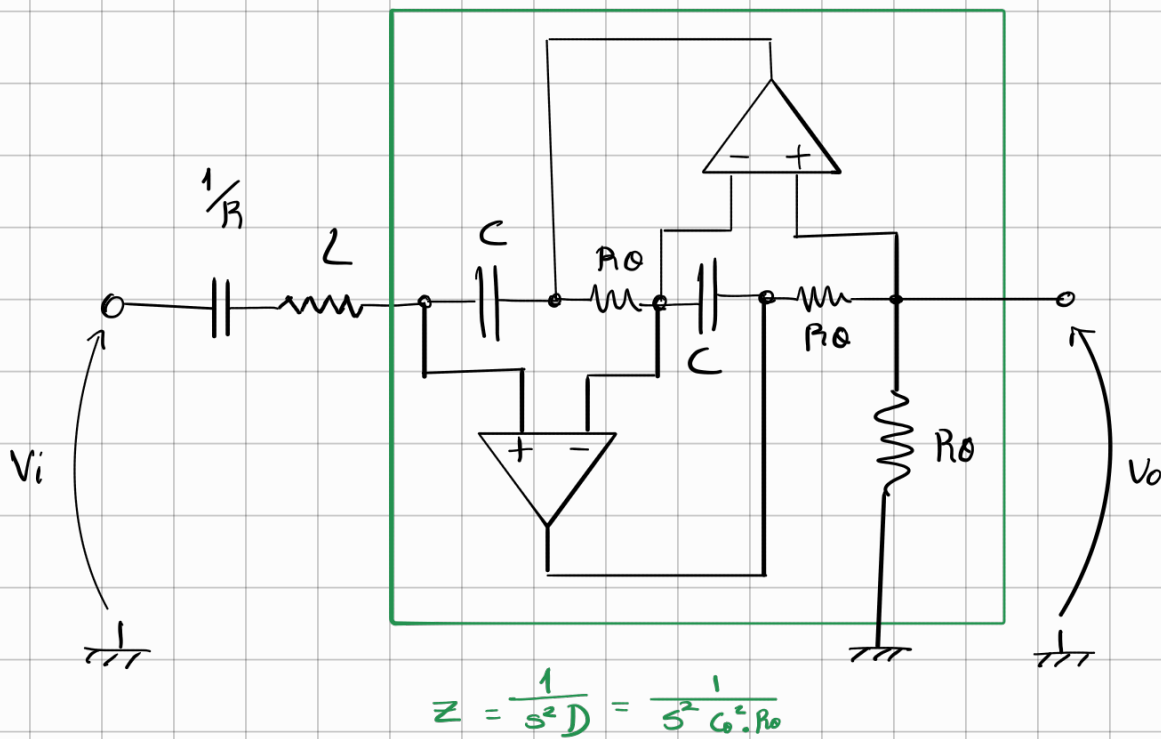
$$\left\{ \begin{array}{l} Q = \frac{1}{CR} = 1,706 \Rightarrow R = \frac{1}{QC} \end{array} \right.$$



Bruton



- El Circuito Eq. aplicando la transf. de Bruton y reemplazando el FDNR:



- $C = C_0^2 R_0$, si $C = C_0$:

$$\Rightarrow R_0 = \frac{1}{C}$$